

# DS 2500 *Fall* 2025 Lab Exercise 05

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**Deadline :** 10/06/2025 11:59 PM EST

**Late Deadline:** 11/03/2025 11:59 PM EST with 50 points off

**Total Points:** 200 pts

**Grading:**

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

**Submission Instructions:**

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

*Disclosure:* We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

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## Allowed packages: requests, BeautifulSoup

Using packages other than `requests` and `BeautifulSoup` will result in 0 point.

### Problem 1 (100pts)

- Task:
  - Scrape the first 3 pages of quotes in the website <http://quotes.toscrape.com>.
  - For each quote, extract:
    - The quote text
    - The author's name
  - Store the extracted data in a dictionary named `author2quotes` where:
    - The author's name as the key
    - A list of quotes by that author as the value
  - Note:
    - Some authors have multiple quotes, so aggregate all quotes under the same author.
    - Apply `.strip()` to all extracted text to remove leading/trailing whitespace.
  - Going to the next page:
    - You can check the URL for the next page and use that information OR
    - The website uses a "Next" button to navigate pages, which you can use that tag to get the additional url.
  - **Hint:**

### Expected Output Format:

```
{"Albert Einstein" : ["The world as we have created it is a process of our
thinking. It cannot be changed without changing our thinking.", ...],
  ...
}
```

### Problem 2 (100 pts)

- URL: <https://catalog.northeastern.edu/course-descriptions/abrd/>
- Task:
  - On this page, each course title follows a format like: **ABRD 5100. International Study–Sweden. (20 Hours)**
  - Scrape all course titles from the page.
  - Extract the country name from each course title (e.g., "Sweden").
    - Use `.split()` extract the country string.
    - Use the predefined list `punctuations` to clean it further.
    - Don't forget to apply `.strip()` to each step to remove leading/trailing whitespace.
  - Collect all unique country names in a list called `countries`.